Adaptive Quantum Error Mitigation Under Fixed Shot Budgets: A Reproducible Simulator Benchmark

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Abstract

Quantum error mitigation (QEM) can reduce bias without encoding a logical qubit, but its auxiliary executions and statistical amplification consume finite sampling resources. This study asks when unmitigated estimation, full-assignment-matrix readout mitigation, and linear zero-noise extrapolation (ZNE) are useful under the same total shot budget. We evaluate $X^{\otimes w}$ on simulated GHZ circuits with widths $w \in \{2, 4, 6\}$, four idealized mixed-noise levels, three shot budgets, and 50 deterministic repeats per applicable operating cell. Among 34 cells for which all three methods completed 50 repeats, readout mitigation attained the lowest root-mean-square error (RMSE) in 21 cells, raw estimation in 9, and linear ZNE in 4. Across completed rows, the respective RMSE values were 0.1647, 0.2793, and 0.2415 for readout, raw, and ZNE. Readout mitigation also failed explicitly in 28 width-6, 1,024-shot rows because the sampled calibration matrix was ill-conditioned. The evidence is entirely analytic or simulated; it does not establish performance on quantum hardware or under calibrated device noise.

1 Introduction

Near-term quantum computations often estimate expectation values from noisy, finite-shot measurements. QEM methods seek to reduce the resulting error without the qubit overhead of full error correction, but they introduce sampling and classical-processing overhead and can amplify variance [6, 4, 2]. Consequently, applying mitigation is not a method-independent prescription: effectiveness depends on the circuit, observable, noise channel, and resource budget.

The research question is: *under a fixed total shot budget, which of raw estimation, readout mitigation, and linear ZNE minimizes estimation error, and where does mitigation fail or waste resources?* The repository implements a simulator-only benchmark with deterministic seeds, repeat-level raw data, explicit resource accounting, and failure statuses. The contribution is an auditable operating-regime comparison, not a claim of new QEM theory or hardware advantage.

Table 1: The four simulated noise levels increase readout and depolarizing error together; none is assigned to a physical device.

Level	p_{01}	p_{10}	p_{1q}	p_{2q}
Zero	0	0	0	0
Low	.005	.004	.001	.010
Medium	.020	.016	.005	.030
High	.050	.040	.010	.080

2 Methodology

2.1 Reference state and observable

For width w, the benchmark prepares

$$|\mathrm{GHZ}_w\rangle = \frac{|0\rangle^{\otimes w} + |1\rangle^{\otimes w}}{\sqrt{2}}, \tag{1}$$

using one Hadamard gate followed by $w - 1$ controlled-NOT gates. The primary observable is $P_w = X^{\otimes w}$, for which the analytic prediction and noiseless statevector result are

$$\mu_{\mathrm{ref}} = \langle \mathrm{GHZ}_w | P_w | \mathrm{GHZ}_w \rangle = 1. \tag{2}$$

Tests cross-check this relation to absolute tolerance 10^{-12} for $w \in \{2, 4, 6\}$.

For N shots with parity eigenvalues $y_s \in \{-1, +1\}$, the raw estimator and plug-in shot-noise standard error are

$$\widehat{\mu} = \frac{1}{N} \sum_{s=1}^{N} y_s, \tag{3}$$

$$\widehat{\sigma}_{\widehat{\mu}} = \sqrt{\frac{1 - \widehat{\mu}^{\,2}}{N}}. \tag{4}$$

2.2 Noise model

Qiskit Aer supplies the noisy target-circuit simulation [5]. The diagnostic mixed-noise model combines independent assignment error, one-qubit depolarizing error, and two-qubit depolarizing error. Table 1 records the exact simulator parameters. These values are controlled study factors, not device calibrations.

2.3 Readout mitigation

For width w, all 2^w computational basis states are calibrated. If $A_{ij} = \Pr(\text{observe } i \mid \text{prepare } j)$ and $\widehat{\boldsymbol{p}}_{\text{obs}}$ is the observed probability vector, the corrected distribution solves

$$\widehat{\boldsymbol{p}} = \arg\min_{\boldsymbol{p} \geq 0} \|A\boldsymbol{p} - \widehat{\boldsymbol{p}}_{\text{obs}}\|_2, \quad \mathbf{1}^{\mathsf{T}}\widehat{\boldsymbol{p}} = 1. \tag{5}$$

This constrained correction follows the assignment-matrix approach [1]. A condition number above 10^8 or a non-finite value produces an explicit `ill_conditioned` status. The total budget is split deterministically: 20% for calibration and 80% for the target. Calibration observations are sampled directly from the declared independent assignment channel with seeded multinomial draws; no hardware calibration data are used.

2.4 Linear zero-noise extrapolation

Global unitary folding constructs

$$U_\lambda = U(U^\dagger U)^{(\lambda-1)/2}, \quad \lambda \in \{1, 3, 5\}, \tag{6}$$

which is ideally equivalent to U while increasing noisy gate exposure. This is a standard digital noise-scaling mechanism [3]. The fixed total shot budget is divided across the three factors. Before the full benchmark, the analysis model was fixed as ordinary least squares,

$$\widehat{\mu}(\lambda) = \beta_0 + \beta_1\lambda, \qquad \widehat{\mu}_{\text{ZNE}} = \beta_0. \tag{7}$$

Extrapolates are retained without clipping and flagged when outside $[-1, 1]$.

2.5 Metrics

For completed repeats $r = 1, \ldots, R$, let $e_r = \widehat{\mu}_r - \mu_{\text{ref}}$. The reported metrics are

$$\text{bias} = R^{-1}\sum_r e_r, \tag{8}$$

$$\text{variance} = (R-1)^{-1}\sum_r (\widehat{\mu}_r - \overline{\mu})^2, \tag{9}$$

$$\text{RMSE} = \sqrt{R^{-1}\sum_r e_r^2}. \tag{10}$$

Method-level summaries are repeat-weighted and conditional on completion. The operating-regime winner is defined only where all three methods have 50 completed repeats, avoiding comparison against a partially failed method.

3 Experiment Setup

The frozen manifest crosses widths $2, 4, 6$; declared depths $1, 3, 6$; four noise levels; shot budgets 1,024, 4,096, and 16,384; three methods; and 50 repeat seeds.

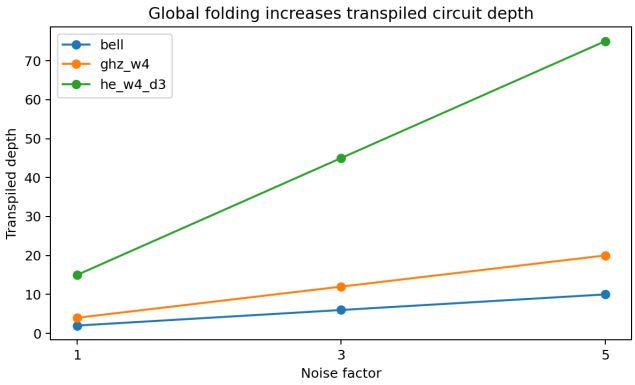


Figure 1: Global folding increases transpiled depth for the tested Bell, GHZ, and hardware-efficient circuits while preserving the ideal unitary in tests.

GHZ preparation is defined only at depth parameter 1, so depths 3 and 6 remain in the manifest as `not_applicable`. The applicable benchmark contains 5,400 repeat-level rows across 36 operating cells and three methods.

All stochastic seeds derive from root seed 20261102 through SHA-256 namespacing. Raw estimation spends all shots on the target. Readout mitigation charges every calibration shot to the same total budget. ZNE charges all three folded-circuit executions to that budget. The environment is Python 3.12 with Qiskit 2.2.3, Qiskit Aer 0.17.2, NumPy 2.3.4, SciPy 1.16.3, and pandas 2.3.3.

Figure 1 verifies that folding increases transpiled depth while tests establish ideal-unitary equivalence to 10^{-12}.

4 Results

4.1 Aggregate error and operating regimes

Figure 2 shows completed-row RMSE by simulated noise. Zero-noise results agree with the analytic reference up to floating-point precision. Beyond zero noise, readout mitigation has the lowest aggregate RMSE: 0.1647 overall, compared with 0.2415 for linear ZNE and 0.2793 for raw estimation. These aggregates do not mean that readout wins every condition.

Figure 3 resolves the aggregate into 36 width–noise–budget conditions. Of the 34 fully completed cells, readout has the lowest RMSE in 21, raw estimation in 9, and linear ZNE in 4. No method dominates the complete operating grid.

4.2 Explicit readout failures

The run retains 5,372 completed and 28 failed rows. All failures occur for width 6, total budget 1,024, and readout mitigation (Table 3). They represent 1.56% of the 1,800 readout rows. With 20% calibration allocation, 204 shots

Table 2: Completed-row bias and RMSE depend on method and simulated noise. Readout has fewer completed rows at medium and high noise because failed rows have no numeric estimate.

Noise	Method	Completed	Bias	RMSE
Zero	Raw	450	0.0000	0.0000
Zero	Readout	450	0.0000	0.0000
Zero	Linear ZNE	450	0.0000	0.0000
Low	Raw	450	-0.0685	0.0759
Low	Readout	450	-0.0465	0.0551
Low	Linear ZNE	450	-0.0416	0.0475
Medium	Raw	450	-0.2261	0.2443
Medium	Readout	447	-0.1246	0.1483
Medium	Linear ZNE	450	-0.1757	0.1927
High	Raw	450	-0.4695	0.4967
High	Readout	425	-0.2563	0.2946
High	Linear ZNE	450	-0.4091	0.4405

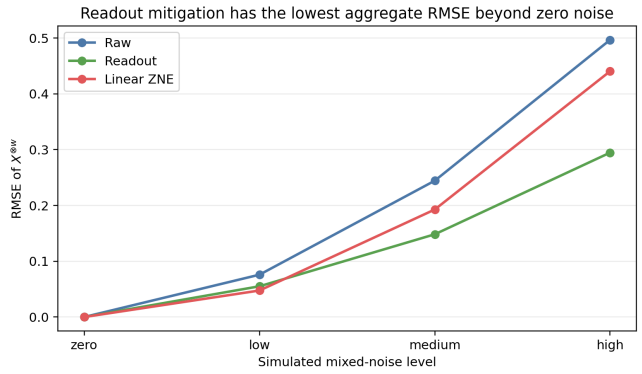


Figure 2: Across completed simulated rows, readout mitigation has lower RMSE than raw estimation and linear ZNE at low, medium, and high mixed-noise levels.

Table 3: All failed simulated rows are explicit conditioning failures at the smallest budget and largest width.

w	Noise	Shots	Method	Count	Status
6	High	1,024	Readout	25	Ill-conditioned assignment matrix
6	Medium	1,024	Readout	3	Ill-conditioned assignment matrix

are divided among 64 basis states, leaving three or four samples per calibration column; the sampled assignment matrix can be singular or exceed the condition-number threshold. This is a supported numerical diagnosis; its hardware relevance remains untested.

4.3 ZNE behavior

The preregistered three-factor linear fit improves aggregate RMSE over raw estimation but wins only 4 complete cells. Only 5 nontrivial extrapolates exceed the Pauli range by more than 10^{-12}; they remain unclipped. Figure 4 shows factor estimates and the fit for the first Day 9 validation repeat. It illustrates the estimator, not the full performance distribution.

5 Conclusion

Under the declared simulator model and equal total shot budgets, QEM effectiveness is operating-regime dependent. Readout mitigation produced the lowest RMSE most often and the lowest aggregate RMSE, but failed when full-matrix calibration was undersampled. Linear ZNE improved aggregate RMSE relative to raw estimation yet was best in only a minority of complete cells. The evidence supports a selector based on width, noise, budget, and method status; it does not support unconditional mitigation.

6 Discussion

6.1 Evidence-supported findings

The comparison demonstrates that mitigation overhead must be charged explicitly, aggregate performance hides regime changes, and failure status is part of method performance. Raw estimation remains preferable in 9 fully completed cells. Omitting the 28 ill-conditioned rows would overstate readout reliability.

6.2 Interpretation and hypotheses

Readout mitigation likely performs well because its calibration uses the same independent assignment channel applied to target measurements. This is an interpretation, not evidence that it remains best under correlated readout, drift, or device calibration error. Linear ZNE may underperform because a first-order model is restrictive and the fixed budget leaves about one third of the

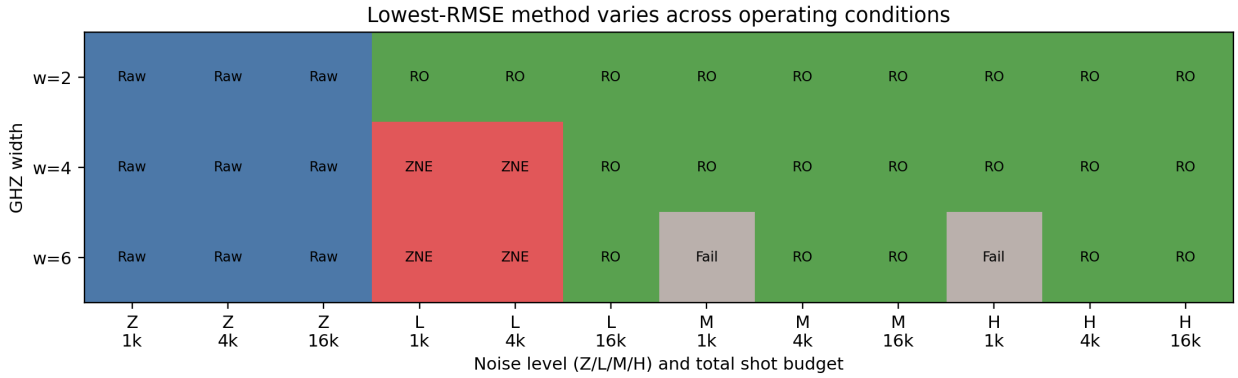


Figure 3: The minimum-RMSE method changes with width, noise, and total shots. Failed conditions are excluded because readout did not complete all 50 repeats.

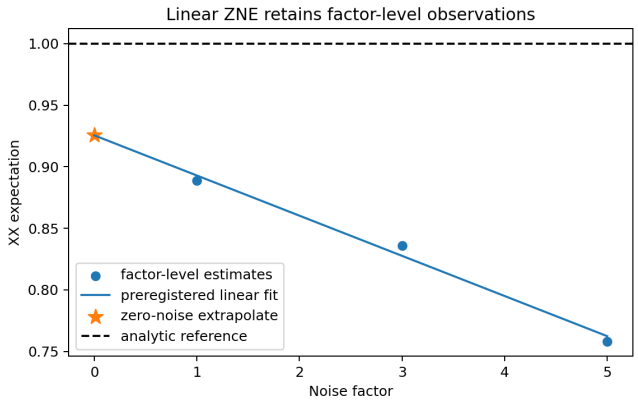


Figure 4: The validation artifact preserves all three noisy estimates and the preregistered extrapolate instead of reporting only a corrected value.

shots at each factor. The benchmark does not isolate these possible causes.

6.3 Limitations

All noisy results are Qiskit Aer simulations with idealized, stationary, Markovian channels. The study contains no hardware-derived or physical experimental data and models no crosstalk, leakage, relaxation, drift, compilation variability, or coherent calibration error. The GHZ benchmark uses only $X^{\otimes w}$ and widths up to six; the declared depth sweep is not applicable. Full-matrix calibration scales as 2^w. Method-level RMSE is conditional on completion, and the operating map contains only 34 complete comparisons. The fixed seed set is not a confidence statement about real devices. No learned or optimized selector is evaluated.

A Reproducibility and Data Contract

```
python -m pytest
python -m src.generate_artifacts
python -m src.generate_days08_12
python -m src.generate_paper_assets
tectonic paper/main.tex --outdir paper
```

Repeat-level observations remain under `results/raw/`; processed coverage is separate. Seeds, statuses, count payloads, and method diagnostics are retained. Wall-clock measurements are excluded from scientific raw rows because they are nondeterministic and are reported separately.

B Analytic and Test Controls

Tests check q0-first bit ordering, seed derivation, Bell and GHZ analytic expectations, statevector agreement, folding equivalence, exact shot accounting, calibration identity, explicit ill-conditioning, ZNE factors, manifest completeness, and deterministic row execution. These validate implementation invariants, not physical fidelity of the noise model.

References

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